Ex.no:4	# EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, conditional filtering, sorting, statistical summary, group aggregation, correlation analysis, and data visualization on the **Tourism Dataset** using Pandas and Matplotlib.

Algorithm:

1. Start.
2. Import the required Python libraries: Pandas and Matplotlib.
3. Load the Tourism dataset into a Pandas DataFrame using pd.read_csv().
4. Inspect the initial records, shape, data types, and numerical summaries using head(), shape, dtypes, and describe().
5. Filter tourism records based on specific conditions such as Rating greater than 4.
6. Filter tourism destinations based on tourism type.
7. Sort the dataset based on Estimated Budget in descending order.
8. Calculate statistical measures such as mean, median, mode, and standard deviation.
9. Group the dataset based on Continent and calculate the average Estimated Budget.
10. Calculate the correlation matrix for numerical attributes.
11. Generate a bar chart for average Estimated Budget by Continent.
12. Generate a scatter plot between Average Duration and Estimated Budget.
13. Stop.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
# ============================================================
# EXPERIMENT NO: 4
# EXPLORATORY DATA ANALYSIS (EDA)
# TOURISM DATASET
# ============================================================


# ------------------------------------------------------------
# 1. IMPORT DATASET
# ------------------------------------------------------------

df = pd.read_csv("tourism_dataset_200.csv")


# ------------------------------------------------------------
# 2. DATASET INSPECTION
# ------------------------------------------------------------

print("===== FIRST FIVE RECORDS =====")
print(df.head())

print("\n===== SHAPE OF DATASET =====")
print(df.shape)
```

```
print("\n===== DATA TYPES =====")
print(df.dtypes)

print("\n===== DATASET INFORMATION =====")
df.info()

print("\n===== SUMMARY STATISTICS =====")
print(df.describe())



# ----------------------------------------------------
# 3. FILTERING DATA
# ----------------------------------------------------

# Tourism destinations with Rating greater than 4

print("\n===== DESTINATIONS WITH RATING > 4 =====")

high_rating = df[df["Rating"] > 4]

print(high_rating.head())



# Tourism destinations with Budget greater than 3000

print("\n===== DESTINATIONS WITH BUDGET > 3000 =====")

high_budget = df[df["Estimated_Budget_USD"] > 3000]

print(high_budget.head())
```

```
# Filter Beach Tourism

print("\n===== BEACH TOURISM DESTINATIONS =====")

beach_tourism = df[df["Tourism_Type"] == "Beach"]

print(beach_tourism.head())


# ----------------------------------------------------
# 4. SORTING DATA
# ----------------------------------------------------

print("\n===== SORT BY ESTIMATED BUDGET (DESCENDING) =====")

sorted_df = df.sort_values(
    by="Estimated_Budget_USD",
    ascending=False
)

print(sorted_df.head())


# ----------------------------------------------------
# 5. STATISTICAL ANALYSIS
# ----------------------------------------------------

print("\n===== ESTIMATED BUDGET STATISTICS =====")
```

```python
print("Mean :",
      df["Estimated_Budget_USD"].mean())

print("Median :",
      df["Estimated_Budget_USD"].median())

print("Mode :")
print(df["Estimated_Budget_USD"].mode())

print("Standard Deviation :",
      df["Estimated_Budget_USD"].std())


# Statistics for Rating

print("\n===== RATING STATISTICS =====")

print("Mean :",
      df["Rating"].mean())

print("Median :",
      df["Rating"].median())

print("Mode :")
print(df["Rating"].mode())

print("Standard Deviation :",
      df["Rating"].std())
```

```
# ----------------------------------------------------
# 6. DESCRIPTIVE STATISTICS
# ----------------------------------------------------

print("\n===== DESCRIPTIVE STATISTICS =====")

print(df.describe())



# ----------------------------------------------------
# 7. GROUPING AND AGGREGATION
# ----------------------------------------------------

print("\n===== AVERAGE ESTIMATED BUDGET BY CONTINENT =====")

continent_avg = df.groupby("Continent")[
    "Estimated_Budget_USD"
].mean()

print(continent_avg)



# Average Rating by Tourism Type

print("\n===== AVERAGE RATING BY TOURISM TYPE =====")

tourism_rating = df.groupby("Tourism_Type")[
    "Rating"
].mean()
```

```
print(tourism_rating)


# ----------------------------------------------------
# 8. CORRELATION ANALYSIS
# ----------------------------------------------------

print("\n===== CORRELATION MATRIX =====")

corr = df[[
    "Average_Duration_Days",
    "Estimated_Budget_USD",
    "Rating",
    "Annual_Visitors"
]].corr()

print(corr)


# ----------------------------------------------------
# 9. VISUALIZATION 1
# ----------------------------------------------------

continent_avg.plot(kind="bar")

plt.title("Average Estimated Budget by Continent")

plt.xlabel("Continent")
```

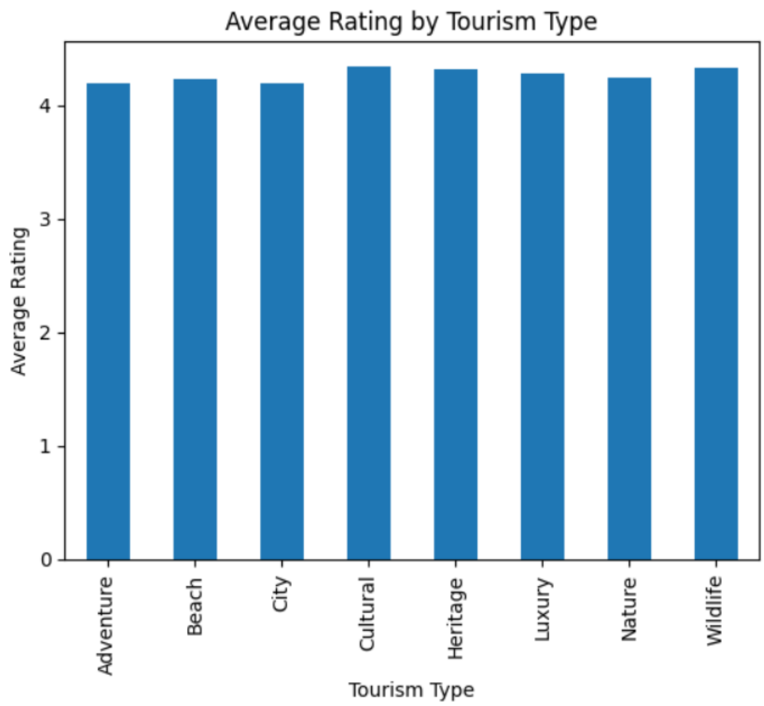
```
plt.ylabel("Average Estimated Budget (USD)")

plt.show()



# ----------------------------------------------------
# 10. VISUALIZATION 2
# ----------------------------------------------------

plt.figure(figsize=(8, 5))

plt.scatter(
    df["Average_Duration_Days"],
    df["Estimated_Budget_USD"]
)

plt.title(
    "Average Duration vs Estimated Budget"
)

plt.xlabel("Average Duration (Days)")

plt.ylabel("Estimated Budget (USD)")

plt.show()



# ----------------------------------------------------
# 11. VISUALIZATION 3
# ----------------------------------------------------
```

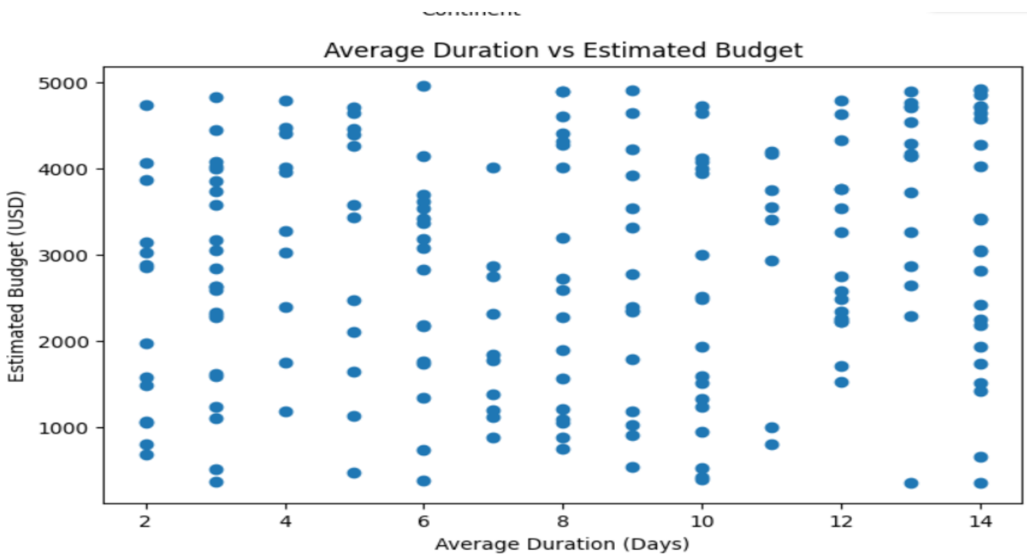
```
tourism_rating.plot(kind="bar")

plt.title("Average Rating by Tourism Type")

plt.xlabel("Tourism Type")

plt.ylabel("Average Rating")

plt.show()



# ------------------------------------------------------
# END OF PROGRAM
# ------------------------------------------------------

print("\n===== EDA COMPLETED SUCCESSFULLY =====")
```

Output:

```
===== FIRST FIVE RECORDS =====
  Tourism_ID  Destination    Country      Continent Tourism_Type Best_Season  \
0       T001  Los Angeles        USA  North America         City      Winter
1       T002    Barcelona      Spain         Europe         City      Winter
2       T003        Paris     France         Europe     Wildlife      Winter
3       T004    Singapore  Singapore           Asia     Wildlife      Spring
4       T005        Tokyo      Japan           Asia     Cultural      Spring

   Average_Duration_Days  Estimated_Budget_USD  Rating  Annual_Visitors  \
0                      3                   504     4.6           133393
1                     11                  3756     3.5            54123
2                     10                  1928     4.6           372696
3                     14                   353     4.6            88707
4                     14                  3057     3.7           204191

  Accommodation_Type Transportation
0              Hotel         Cruise
1          Apartment         Cruise
2           Homestay         Cruise
3             Hostel          Train
4             Hostel         Cruise

===== SHAPE OF DATASET =====
(200, 12)

===== DATA TYPES =====
```

```
===== DATA TYPES =====
Tourism_ID                object
Destination               object
Country                   object
Continent                 object
Tourism_Type              object
Best_Season               object
Average_Duration_Days      int64
Estimated_Budget_USD       int64
Rating                   float64
Annual_Visitors            int64
Accommodation_Type        object
Transportation            object
dtype: object

===== DATASET INFORMATION =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Tourism_ID             200 non-null    object 
 1   Destination            200 non-null    object 
 2   Country                200 non-null    object 
 3   Continent              200 non-null    object 
 4   Tourism_Type           200 non-null    object 
 5   Best_Season            200 non-null    object 
 6   Average_Duration_Days  200 non-null    int64  
 7   Estimated_Budget_USD   200 non-null    int64  
 8   Rating                 200 non-null    float64
 9   Annual_Visitors        200 non-null    int64  
 10  Accommodation_Type     200 non-null    object 
```





Result:

Thus, **Exploratory Data Analysis (EDA) was successfully performed on the Tourism Dataset using Pandas and Matplotlib**. The dataset was inspected, filtered, sorted, and analyzed using statistical measures. Grouping and aggregation were performed, correlation between numerical variables was calculated, and the results were visualized using bar charts and scatter plots.